Optical parametric oscillators and coherent Ising machines

Nicholas Chancellor

October 18, 2022

October 18, 2022

Disclaimer: lecture notes may contain "typo" type errors (factors of 2, minus signs, etc...) if something like this looks wrong there is a good chance it is, please let me know

These notes are based partially on "Nonlinear Optics" by Robert Boyd ISBN: 978-0-12-369470-6, available for purchase at[1]: https://www.elsevier.com/books/nonlinear-optics/boyd/978-0-12-369470-6 as well as Gerry and Knight, coherent Ising machine discussion based on papers especially Yamamoto et. al. npj Quantum Information volume 3, Article number: 49 (2017) https://doi.org/10.1038/s41534-017-0048-9.

1 Motivation

- Optical parametric oscillators (and amplifiers) are important components in many quantum optics systems, it is therefore worth devoting a whole set of notes to them
- Some of the ideas around thresholds in these systems are transferable to many others such as lasers
- Coherent Ising machines are a model of quantum[2] computation which uses photons to solve problems stated as Ising models, if we don't explain carefully, people will just assume that the EQC is a coherent Ising machine, to show the difference we need to know how a coherent Ising machine works.
- The way in which the mode of operation of the large scale coherent Ising machines currently in existence is not meaningfully quantum is a useful study in what to avoid if you want to make a truly quantum computer[3]

[1] while this is a good book in general I don't necessarily think that everyone needs a copy of it the way we all have a copy of G+K (get QCI to buy you one if you don't already have it), maybe only get it if you want to know how the experiments work

[2] or classical in many of the real implementations, explained later in the notes

[3] This goal shouldn't be assumed always, it could be interesting (and profitable) to build unconventional *classical* computers if they can improve computation in a meaningful way

- There is significant similarity in the structure of a coherent Ising machine and the RPC, although the purposes are quite different

2 Optical parametric amplification

- Consider a non-linear crystal with the following non-linear Hamiltonian [4]

$$\hat{H} = \chi^{(2)} \left(\hat{a}_p \hat{a}_s^\dagger \hat{a}_i^\dagger + \hat{a}_p^\dagger \hat{a}_s \hat{a}_i \right) \tag{1}$$

- $\chi^{(2)}$ is usually small in practice, but if a large number of photons are already present, say in mode p, than the overall contribution can be significant, since the effective strength the other two modes see will scale as $\sqrt{n_p}$

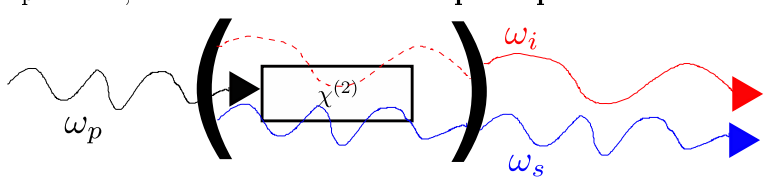
- When acting on a number state $\hat{a}_p \hat{a}_s^\dagger \hat{a}_i^\dagger$ will give a matrix element proportional to $\sqrt{n_p(n_s+1)(n_i+1)}$, therefore more photons in mode i mean that more photons will be created in mode i (and s)

- Lets consider the case where $n_p \gg n_s$ and $n_p \gg n_i$ since the square root function is concave, than for any $n > m \geq 1$ $\sqrt{n-1}\sqrt{m+1} > \sqrt{n}\sqrt{m}$, therefore even if the photons in mode s quickly leave the material, the action of $\hat{a}_p \hat{a}_s^\dagger \hat{a}_i^\dagger$ will lead to faster creation of photons in i and s (and therefore absorption of p), of course this effect is even more dramatic if both are retained

- Effectively the crystal can amplify an optical signal in mode i and create a matching signal in mode s, and as long as the p is not too depleted this effect gets stronger the longer the light travels through the material

- If we assume that the absorption of photons from p can be ignored, and both s and i remain in the material, than the number of photons in s and i will grow exponentially as the photons travel

- This effect can be made even stronger by adding a cavity which causes photons in mode i (or even both i and s) to pass through the medium multiple times, this device is known as an **optical parametric oscillator**



[4] s, p and i stand for "pump", "signal", and "idler" respectively, the reason for this terminology is mostly historical, but it is widely used (including G+K), so I will use it to, Boyd tends to use 1,2,3

3 Optical parametric oscillator (OPO)

- There are three types of optical parametric oscillators often considered
  1. Singly resonant: either mode s or mode i, are significantly trapped by the cavity, but not both
  2. Doubly resonant: both s and i are significantly trapped by the cavity, but they are both distinct modes
  3. Degenerate: s and i are identical modes (i.e. the terms become $\left(\hat{a}_p\hat{a}_s^\dagger\hat{a}_s^\dagger + \hat{a}_p^\dagger\hat{a}_s\hat{a}_s\right)$) and significantly trapped by the cavity
- In all of these cases, there is an interesting tipping point: the point where photons in s and i are created faster than they are lost to the cavity, this is known as a **threshold**, below the threshold the OPO will act as an amplifier, above the threshold, the OPO will spontaneously produce large numbers of s and i photons, even if these modes are initially empty
- Boyd shows how to calculate the threshold in chapter 2.9 (using classical electromagnetism), but the details are probably not important to us so much as the fact that it exists
- Different applications in different regimes
  - Below threshold: Amplifier, light can be input in one mode and is amplified into the same mode, or potentially a different mode
  - Above threshold: Basically an optically pumped laser, can produce light at some frequency when pumped at a different frequency
  - Around/crossing threshold: coherent Ising machine
- Which mode is produced by OPO (assuming above threshold) is determined by length of cavity (light will stay in cavity much longer if **resonant**, in other words if it can "fit" such that there are nodes at each side of the cavity), as well as the properties of the pumping and non-linear material
  - Once one mode reaches threshold the pump is depleted, so it only picks one (this is important for how a coherent Ising machine works)
  - Figuring out which mode is produced can be complicated in doubly resonant case, singly resonant often used for this reason
  - If there are multiple wave packets, determining how threshold is crossed can even be an NP-hard (Ising) problem; this is the principle behind a coherent Ising machine

4 Phase relationship between signal and pump (degenerate OPO)

- Let's consider the degenerate case, the action of the non-linear crystal is then $\left(\hat{a}_p\hat{a}_s^\dagger\hat{a}_s^\dagger + \hat{a}_p^\dagger\hat{a}_s\hat{a}_s\right)$
- If we assume that the pump mode p is not depleted than the action can be approximated as the squeezing operation

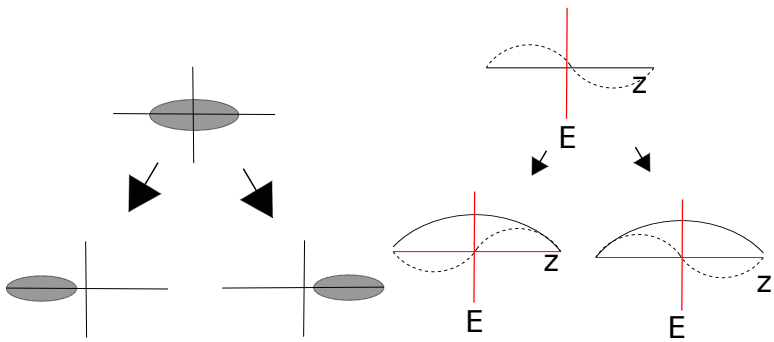
$$\exp\left[\xi\hat{a}_s\hat{a}_s + \xi^\star\hat{a}_s^\dagger\hat{a}_s^\dagger\right] \tag{2}$$

- If we are below threshold and only apply light in the pump mode, our OPO will produce a squeezed vacuum
- Above threshold, we still get squeezing, but also get non-zero electric field expectation, but what modes are allowed?
  - Can get a hint by applying $\exp\left[\xi\hat{a}_s\hat{a}_s + \xi^\star\hat{a}_s^\dagger\hat{a}_s^\dagger\right]$ to a coherent state $|\alpha\rangle = \exp(-\frac{1}{2}|\alpha|)\sum_n \frac{\alpha^n}{n!}|n\rangle$
  - Let us consider the effect of phases $|\alpha\rangle = |e^{i\phi}|\alpha|\rangle$, if we apply the first order Taylor expansion$1+\xi\hat{a}_s\hat{a}_s+\xi^\star\hat{a}_s^\dagger\hat{a}_s^\dagger$, the coefficient for state $|n\rangle$ (assuming $n \geq 2$) will be[5]

$$\exp(-\frac{1}{2}|\alpha|)\frac{|\alpha|^n}{n!}\exp(in\phi)\left(1+\xi|\alpha|^2\exp(2i\phi)+\xi|\alpha|^{-2}\exp(-2i\phi)\right) \tag{3}$$

  - We therefore get constructive interference from applying the OPO if $2\phi = 2m\pi$ where m is an integer, or if $\phi = 0$ or $\phi = \pi$, in other words if α is real (but can be either positive or negative)
  - Same argument holds after squeezing, coefficients will remain positive
  - System is therefore bistable, will produce electromagnetic waves with one of two phase relationships between p and s
- This can also be understood classically, by frequency matching, the signal wave will be twice the wavelength of the pump so therefore too distinct ways to match the node locations, either with the slope of the fields initially in the same versus opposite directions
- Two ways to visualize, phase space (left) and wave matching (right), here the dashed line indicates the field from the pump beam

[5] note that I have assumed here that ξ is real and therefore $\xi^\star = \xi$



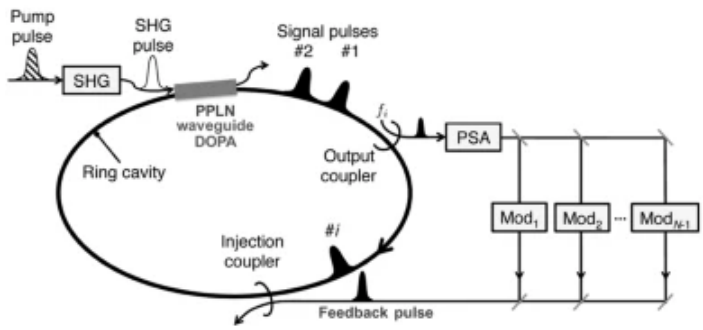
- But which happens when we cross threshold? If we don't do anything than random fluctuations will select one over the other giving a random result (spontaneous symmetry breaking), if we plot the average over many runs the mean value of the phase will look like a pitchfork, hence this is sometimes called a "pitchfork" bifrication
  - But we can also seed a small amount of light with either the + or − phase, the device will amplify that input, using this feedback between multiple modes is the basic operating principle of a coherent Ising machine

5 Building a coherent Ising machine

- A lot of discussion here taken from Yamamoto et. al. npj Quantum Information volume 3, Article number: 49 (2017) https://doi.org/10.1038/s41534-017-0048-9
- If our pump beam consists of spatially-separated wave packets, than the OPO will produce multiple wave packets, each with their own phase relationship to the pump beam → phase of each effectively acts as a binary (i.e. Ising) variable above threshold
- Interaction can be achieved by mixing light from one packet into others, possibly amplifying and modifying the phase (to get anti-ferromagnetic)
- Threshold will now depend on combination of modes due to feedback, minimum energy configuration will be the first to pass threshold
- In principle (fully quantum version):
  - Everything I describe above can be done with delay lines and other optical components
  - If done this way, a coherent Ising machine is a fully quantum model of computing, with no intermediate measurements and full entanglement possible between packets

- Similar to quantum annealing, except for the fact that more has been proven for quantum annealing (there is an annealing version of the Grover problem with an analogous speedup, I don't think the same exists for coherent Ising[6])
- This has been done on a small scale (4 variables in Yamamoto et. al.)

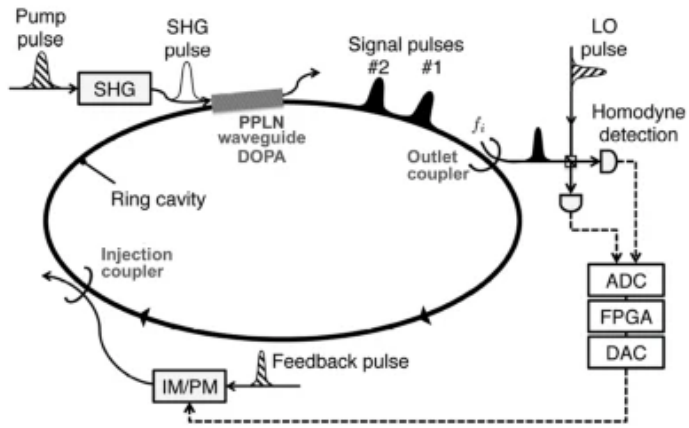
- Diagram of what this looks like (taken from Yamamoto et. al.)



- In practice (classical version):
  - Engineering a large number of interactions using delay lines is not practical from an engineering standpoint
  - But it is possible to measure and create feedback which approximates what delay lines would have given
  - This allows coherence locally, but not entanglement between the packets, still potentially very powerful , but not quantum computing in the way we usually imagine it
    - In particular, lack of coherence between wave packets rules out any kind of Grover type speedup, wavepackets are effectively being used as anlog components within a larger classical computer
    - Having individual quantum components does not make a device a quantum computer, transistors rely on quantum effects to operate but we wouldn't call a laptop or smartphone a quantum computer
  - These can be large, and can perform pretty well in practice

- Diagram of what this looks like (taken from Yamamoto et. al.)

[6] There is theoretical evidence for quantum correlations and entanglement, see https://arxiv.org/abs/1506.00135



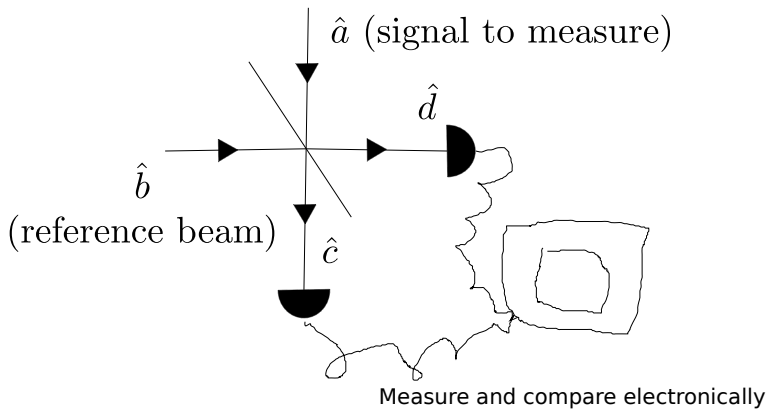
6 How to measure out the state of each variable: Homodyne detection

- Phase is only defined relative to the local oscillator, therefore to measure out the Ising variables we must use a process known as Homodyne detection (discussed in section 7.3 in G+K)
- Assume we have a strong light signal at the frequency $\omega_s = \frac{\omega_p}{2}$ produced as part of our laser and in phase with the pump beam (one of the two possible options) if we didn't have this, we could make it with another OPO, it would be squeezed in that case, but this wouldn't affect the basic argument
- Now consider that we send these two pulses into a $50:50$ beam splitter, if we use $\hat{a}$ for the light we want to measure and $\hat{b}$ for the reference, we see that the difference in intensity on each arm of the beamsplitter is[7]

$$I_c - I_d \propto \langle \hat{n}_c - \hat{n}_d \rangle = \langle \hat{c}^\dagger \hat{c} - \hat{d}^\dagger \hat{d} \rangle = \tag{4}$$

$$\frac{1}{2}\langle (\hat{a}^\dagger - i\hat{b}^\dagger)(\hat{a} + i\hat{b}) - (\hat{b}^\dagger - i\hat{a}^\dagger)(\hat{b} + i\hat{a}) \rangle = i\langle \hat{a}^\dagger \hat{b} - \hat{a}\hat{b}^\dagger \rangle \tag{5}$$

[7] The factor of i which appears in the last expression initially appears alarming, since this is a physical quantity and therefore its expectation should be a real number, however, the quantity within the brackets is **anti-**Hermitian and therefore will have purely imaginary expectation values, leading to an overall real number



- If the field associated with $\hat{b}$ is a coherent state described by $|\beta \exp(i\omega_s t)\rangle = ||\beta| \exp(-i\omega_s t + i(\theta - \frac{\pi}{2}))\rangle$, then this expression reduces to something proportional to $\langle \hat{a}^\dagger \exp(i\omega_s t - i\theta) + \hat{a} \exp(-i\omega_s t + i\theta)\rangle$
- We now observe that $\langle \hat{a} \rangle$ is also time dependent $\langle \hat{a} \rangle = \langle \hat{a}_0 \rangle \exp(i\omega_s t)$, plugging this in gives $\langle \hat{a}_0^\dagger \exp(-i\theta) + \hat{a}_0 \exp(i\theta)\rangle$, which is exactly a measure of the expectation of the quadrature operator along a direction defined by θ, we can therefore distinguish the states of the Ising variables based on which detector sees more intense light
- Note that G+K focuses on using this method to measure squeezing by measuring fluctuations in the intensity difference, but can also directly measure the quadrature